

Tkinter – Complete Guideline for Python GUI

1. What is Tkinter?

Tkinter is Python's standard GUI library used to create desktop applications. It comes pre-installed with Python (no need to install manually).

Import Syntax:

- `import tkinter as tk`
`from tkinter import messagebox`

2. Your First Tkinter Window

Example code to create a simple window:

`import tkinter as tk`

```
root = tk.Tk()
root.title('My First Tkinter App')
root.geometry('400x300')
```

```
label = tk.Label(root, text='Hello, Tkinter!', font=('Arial', 16))
label.pack(pady=20)
```

```
root.mainloop()
```

3. Common Tkinter Widgets

List of commonly used widgets:

Widget	Description	Example
Label	Displays text	<code>tk.Label(root, text='Welcome')</code>
Button	Clickable button	<code>tk.Button(root, text='Click Me')</code>
Entry	Input box	<code>tk.Entry(root)</code>
Checkbutton	Checkbox	<code>tk.Checkbutton(root, text='Accept')</code>
Radiobutton	Radio option	<code>tk.Radiobutton(root, text='Male')</code>
Listbox	List of items	<code>tk.Listbox(root)</code>
Frame	Container for widgets	<code>tk.Frame(root)</code>

4. Geometry Managers

Tkinter provides three layout managers: `pack()`, `grid()`, and `place()`.

5. Handling Button Clicks

You can assign a function to a button using the `command` property.

```
def greet():  
    print('Hello, User!')  
  
button = tk.Button(root, text='Greet', command=greet)  
button.pack(pady=10)
```

6. Taking Input and Showing Output

```
def show_name():  
    name = name_entry.get()  
    messagebox.showinfo('Welcome', f'Hello {name}!')
```

7. Message Boxes

Examples of popup dialogs:

```
messagebox.showinfo('Info', 'This is info')  
messagebox.showwarning('Warning', 'This is warning')  
messagebox.showerror('Error', 'This is error')
```

8. Changing Styles

```
label = tk.Label(root, text='Stylish Text', fg='white', bg='blue', font=('Arial', 16, 'bold'))
```

9. Multiple Windows / Navigation

Use `Toplevel()` to open a new window without closing the current one.

10. Frames

Frames are like sections or divs to organize widgets together.

11. Closing the App

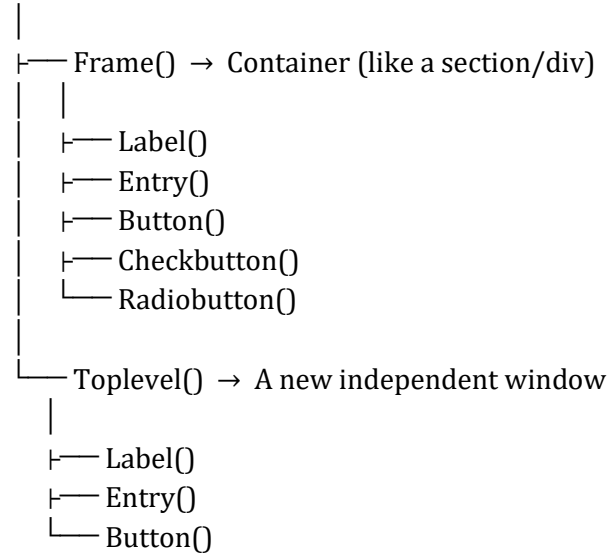
`root.destroy()` closes the window.

12. Tkinter + Database

You can integrate Tkinter with MySQL to fetch or store data dynamically.

Tkinter Widget Hierarchy

Tk() → The Main Window (Root)



ATM Layout Example (Hierarchy)

Tk() → "ATM Login"

